

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)
ON

Civis - Smart Civic Issue Reporting Platform

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “Title of the Project” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of <Name of the Mentor>. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

(Varun Bali)

Date: 6 May, 2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

Nowadays urban areas are experiencing many civic issues such as water leakage, damaged streetlights and unmanaged waste which leads to environmental pollution. As complaint processes are slow, more time consuming and lack of clarity which leads citizens to lose trust in local authorities. To solve the issues this project aims at a **Smart Civic Issue Reporting Platform** that enables people to report issues using their mobile or any electronic devices with visual evidence on our platform. The system verifies complaints by using GPS for sharing your current location to check validity of documents. Authorities receive verified reports by using one screen to monitor and managing everything easily. Our platform is created using two major languages one is react and other is spring boot which is for front-end and back- end simultaneously. So, our project came forward with an idea that you should have to upload an image for evidence to reduce fake reporting.

1.1 Problem Background

Urban communities nowadays are facing many civic problems such as damaged roads, streetlights issues, water supply problems, garbage bins not being cleaned and many more infrastructure related issues. The methods we use for solving civic issues is usually manual and inefficient. Citizens may have to visit government offices, make phone calls for the complaints which are often being ignored or go unrecognized. as still there is no proper solution to track or check the false reports. On the other hand, there is no way for the citizens to check if their complaints are processed or resolved.

On the other side, municipal authorities struggle to manage complaints due to system failure. Delays in communication, documentation further slow down the resolution process.

Therefore, there is a strong need for an online system for citizens to report civic issues accurately with proof and location details, while giving authorities a proper dashboard to manage and to resolve complaints efficiently. As it reduces fake reports and gets more people involved.

1.2 Objective of the Project

- To provide a complaint tracking system for citizens to check the status and progress of their submitted complaints.
- To create an admin management system for authorities to review and check complaints systemically.
- To improve relation among municipal authorities with people through centralized complaint handling process
- By maintaining complaint history, status updates and records of action taken will make the system work more clearly.
- To create an online website to enable citizens to report civic issues in a simple and efficient manner.

1.3 Proposed Solution / System Overview

We come forward with a solution that is digital civic which is basically used for complaints and resolving public issues. The system provides a common online tool for the people where they can register securely, submit complaints and follow the progress of their request in an organized way . By digitizing the complaint process, the project doesn't need any manual reporting methods and makes issue management more clear and managed.

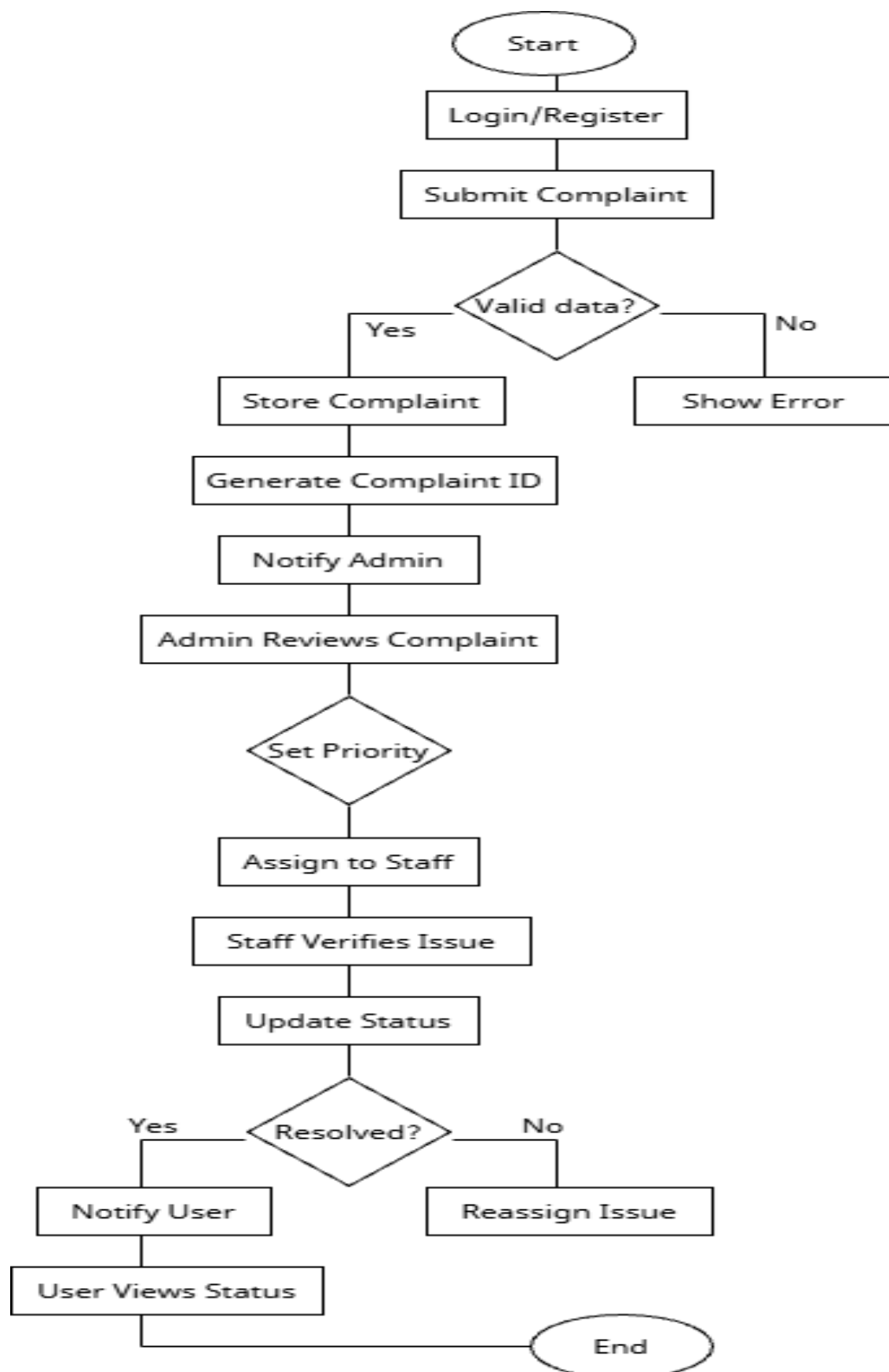
The system is built around two main user roles: citizens and administrators. Citizens can submit complaints by entering issue information, description, location. Each complaint is saved with a different identifier so that it can be tracked. The citizens later can view their complaints and check their status, which helps them to stay updated.

On the administrative side, the system provides a management interface through which authorized officials can access complaint records, examine submitted issues as the resolution process moves forward.

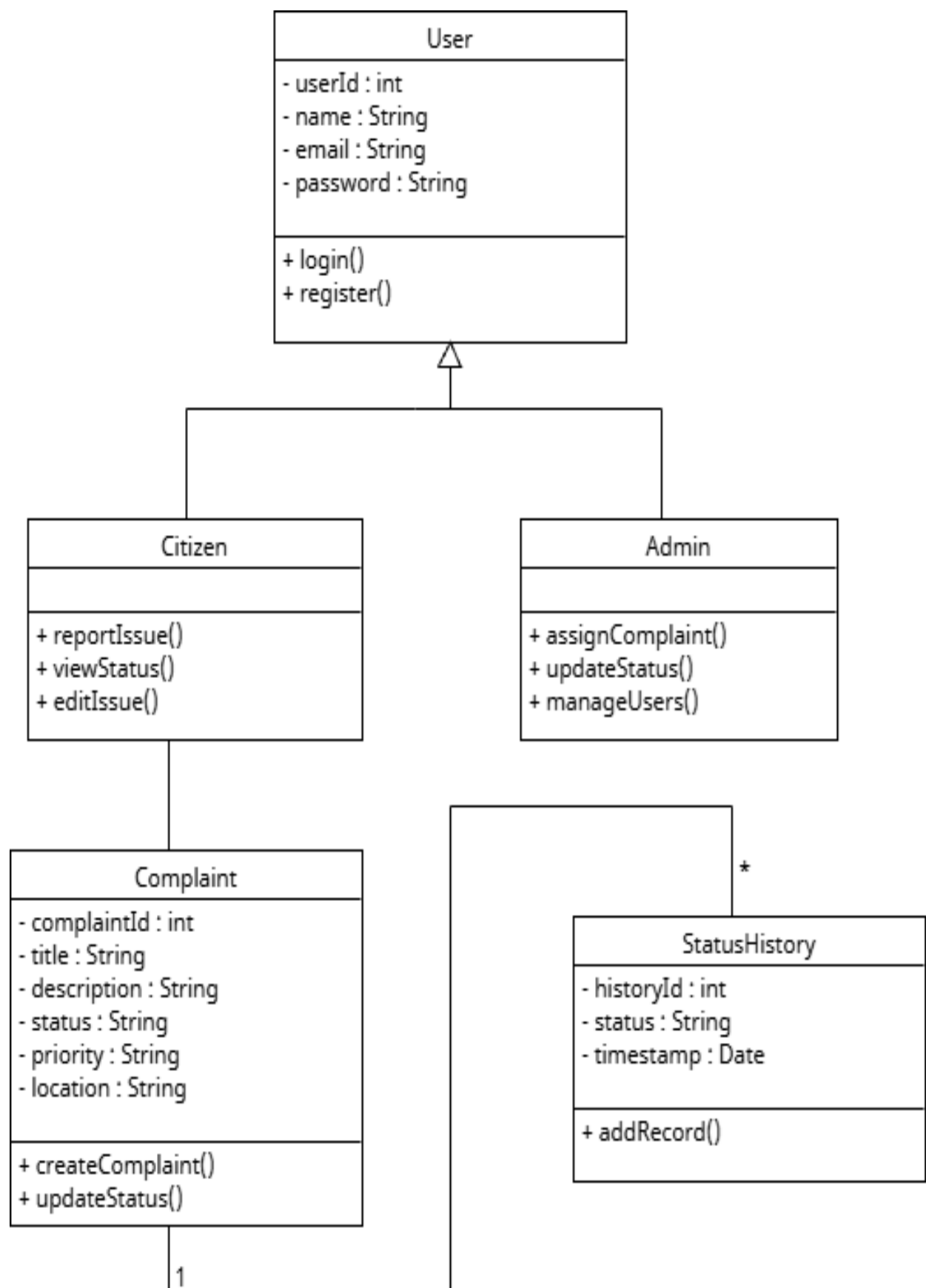
Every important step, including status changes, is recorded in the complaint history, which improves accountability and works smoothly.

Overall, the project offers a centralized complaint registration and monitoring. The purpose behind this is to improve coordination among the people and government officials, support faster handling of civic problems to build stronger and trustful processes.

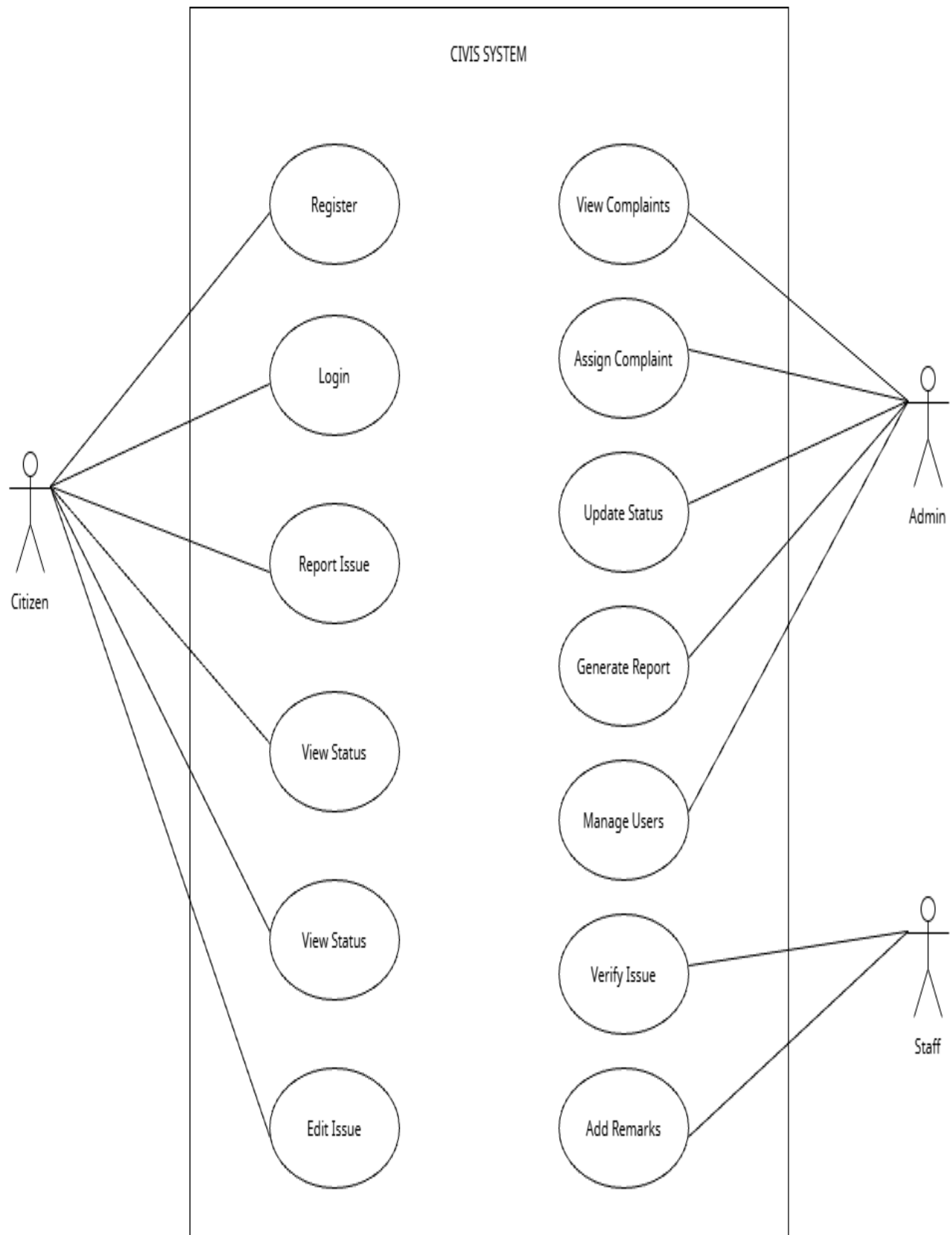
1.3.1 Platform Workflow



1.3.2 Class Diagram



1.3.3 Use-Case Diagram



1.4 Key Features and Functionality

The project offers a range of essential features and efficient management of civic complaints. It begins with a safe login system, where citizens can use through their verified passwords. Each complaint is linked to a verified person and makes the system more dependable.

A primary function is the complaint submission module. Users can share their problems by giving full information. Every submitted complaint has its own verified ID so that officials can easily track it.

This system also includes a detailed tracking feature. Users can view all tracking complaints they have submitted and check their status at any time. Since it maintains a complete timeline of updates related to each complaint, citizens receive a clear view of the process being made.

For the authorities it provides a complete solution to the admin panel. Authorized officials can review all registered complaints, adjust priorities, update status and manage resolution steps.

This system keeps the past data safe so that it runs smoothly. Every change is recorded, ensuring that both citizens and administration have access to a traceable and well-organized complaint management process.

1.5 Technology / Method Used

Technology	Purpose
React.js	Powers the entire user interface through a modular, component-based approach, enabling smooth page transitions, real-time state updates, and a responsive experience across all devices.
TypeScript	Used in the frontend to add type safety and improve code reliability.
Vite	Used as the frontend build tool and development server for fast project setup and execution.

React Router DOM	Used for navigation between pages such as login, dashboard, complaint reporting, and admin dashboard.
Leaflet	Used to provide map-based location selection while submitting complaints.

Java	Used as the main programming language for backend development.
Spring Boot	Used to develop the backend application and REST APIs for complaint management and authentication.
Spring Security	Used to secure the application and control access to protected user and admin endpoints.
Spring Data JPA	Used to simplify database access and perform CRUD operations on application data.
Hibernate	Used as the ORM framework to map Java classes to database tables.
H2 Database	Used as the default in-memory database for local development and testing.
PostgreSQL	Used as the persistent production database for storing users, complaints, and related records.
JWT	Used to generate and validate authentication tokens for secure user sessions.
Twilio	Used to send OTP messages for mobile-based registration and login.
Gradle	Used to manage backend dependencies and build the Spring Boot application.

1.6 Expected Outcome and Benefits

The project is expected to deliver a digital system that makes work simple and systemizes the reporting and management of civic issues. The system will enable citizens to lodge complaints effortlessly, view their status and stay updated with responsible authorities. Thus creates a more accessible and well planned for handling public complaints.

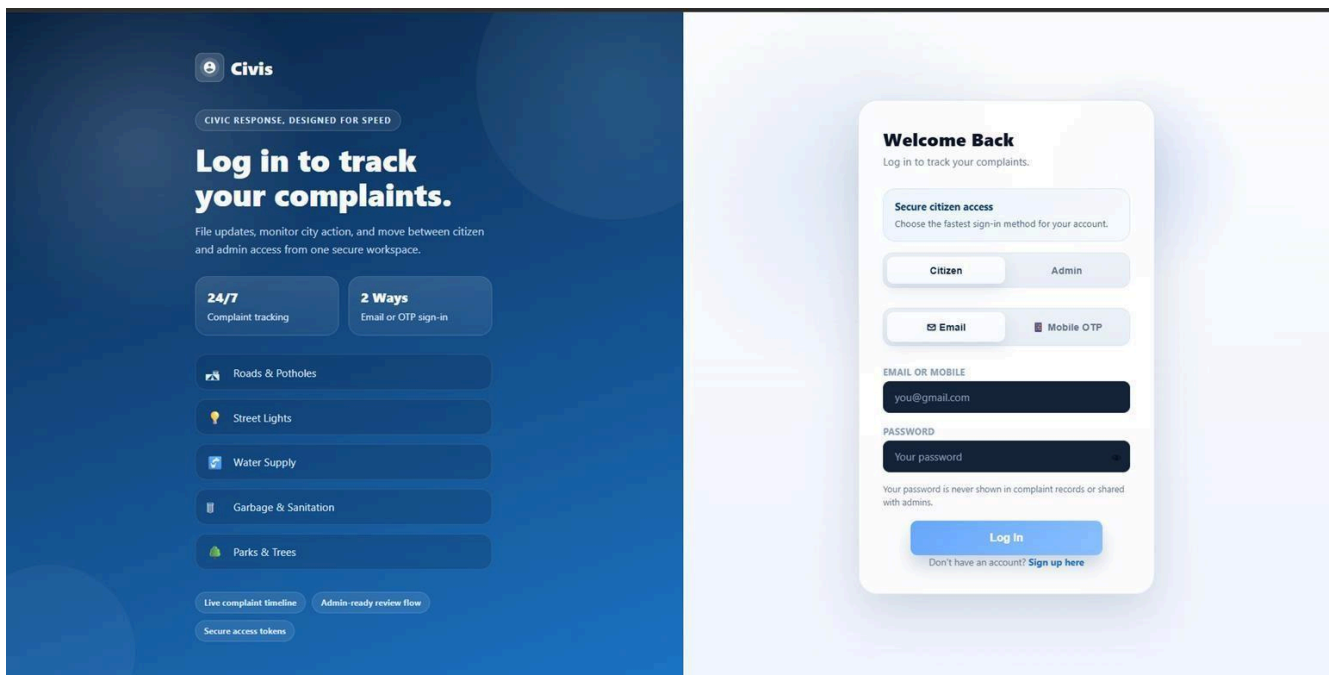
A key benefit of the system is the enhanced benefit provided to the public. As now the public doesn't have to wait for filling complaints they can submit on an online portal and follow their progress. Due to this both citizens and officials valuable time is saved and confusion also.

From the administrative perspective, this system's motive is to improve operational efficiency, officials working can check the incoming complaints and update status changes. It helps to solve civic problems more quickly and smoothly.

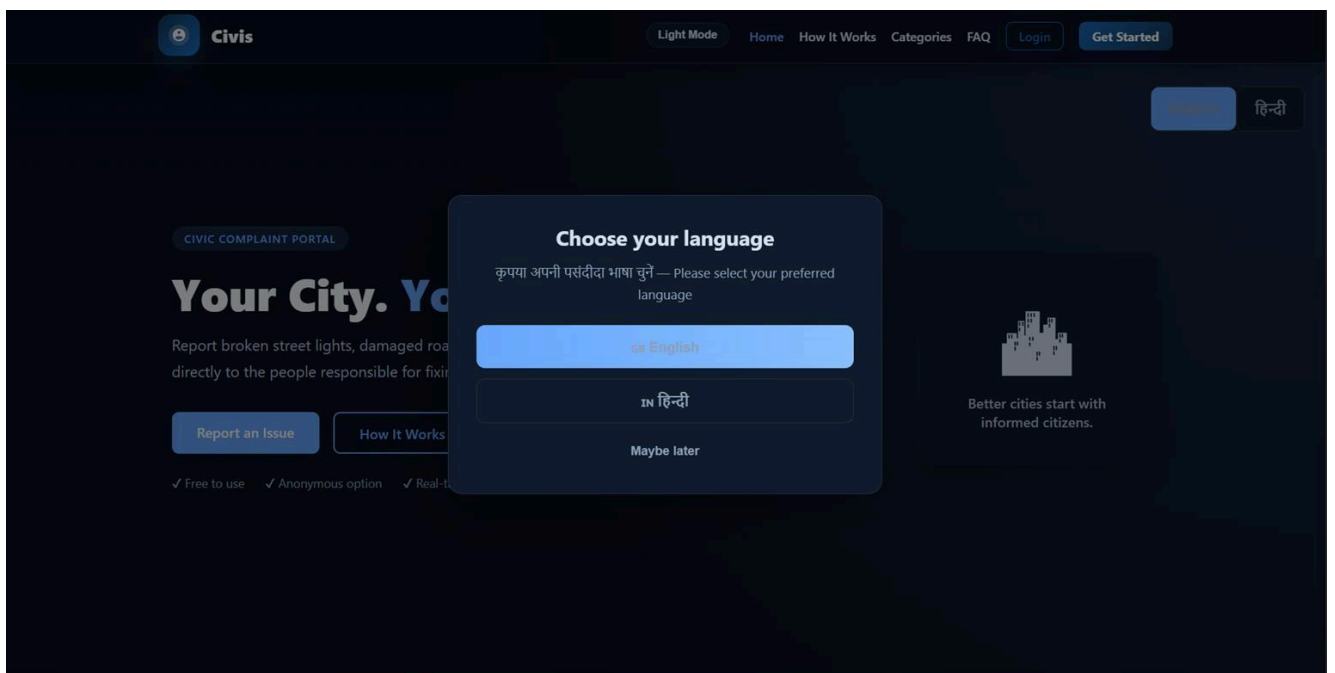
The project also supports greater transparency and responsibility. The complaints update, action taken and resolution detail is stored within the department so that the system remains transferable and safe. As this builds the confidence between the public and officials.

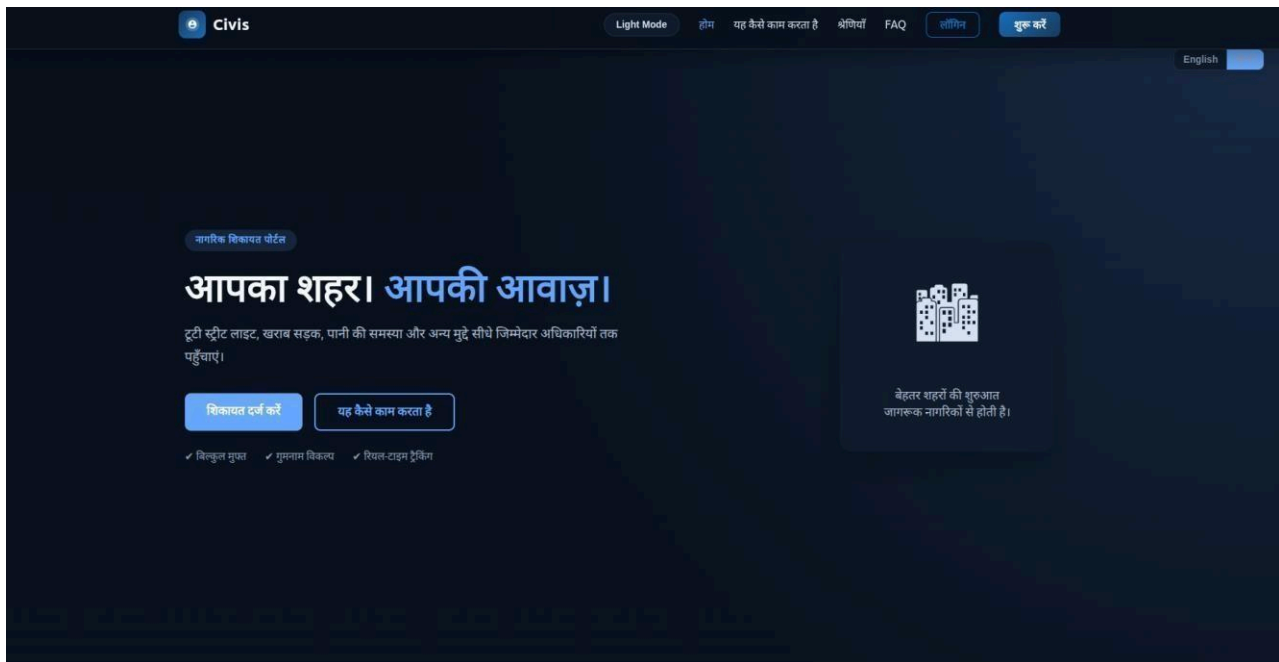
This system is expected to improve interaction and speed up problem solving issues this helps in making management and community organized.

1.6 Login/Signup Page

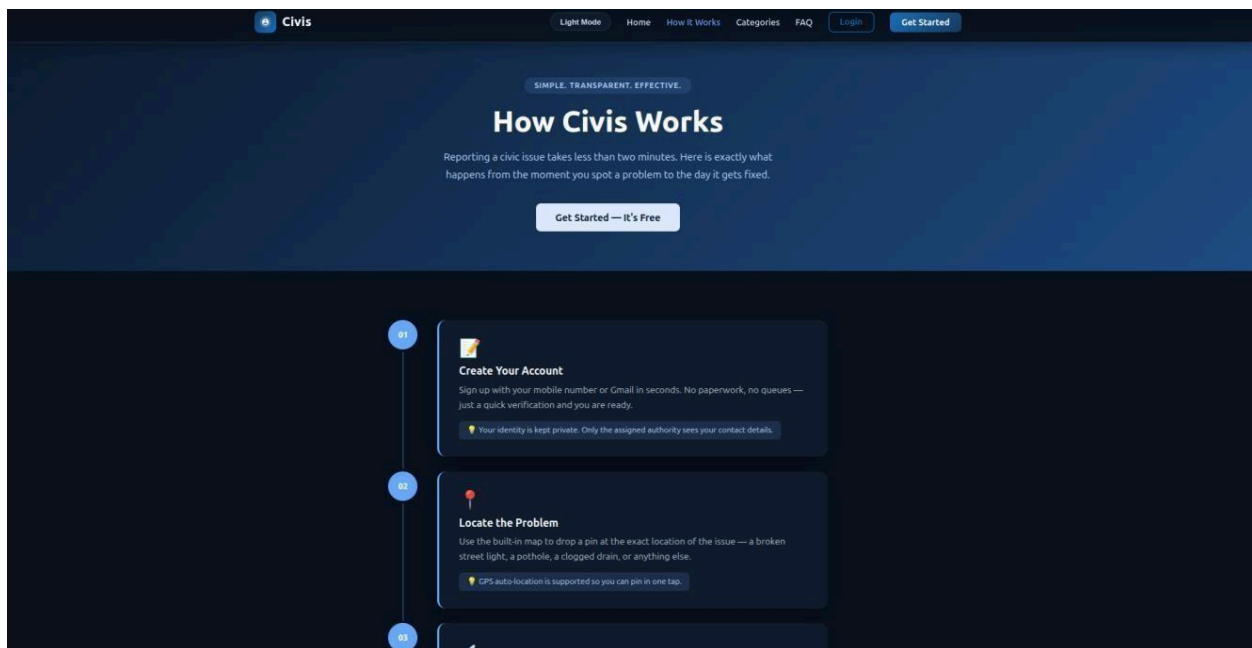


1.6.1 Landing Page

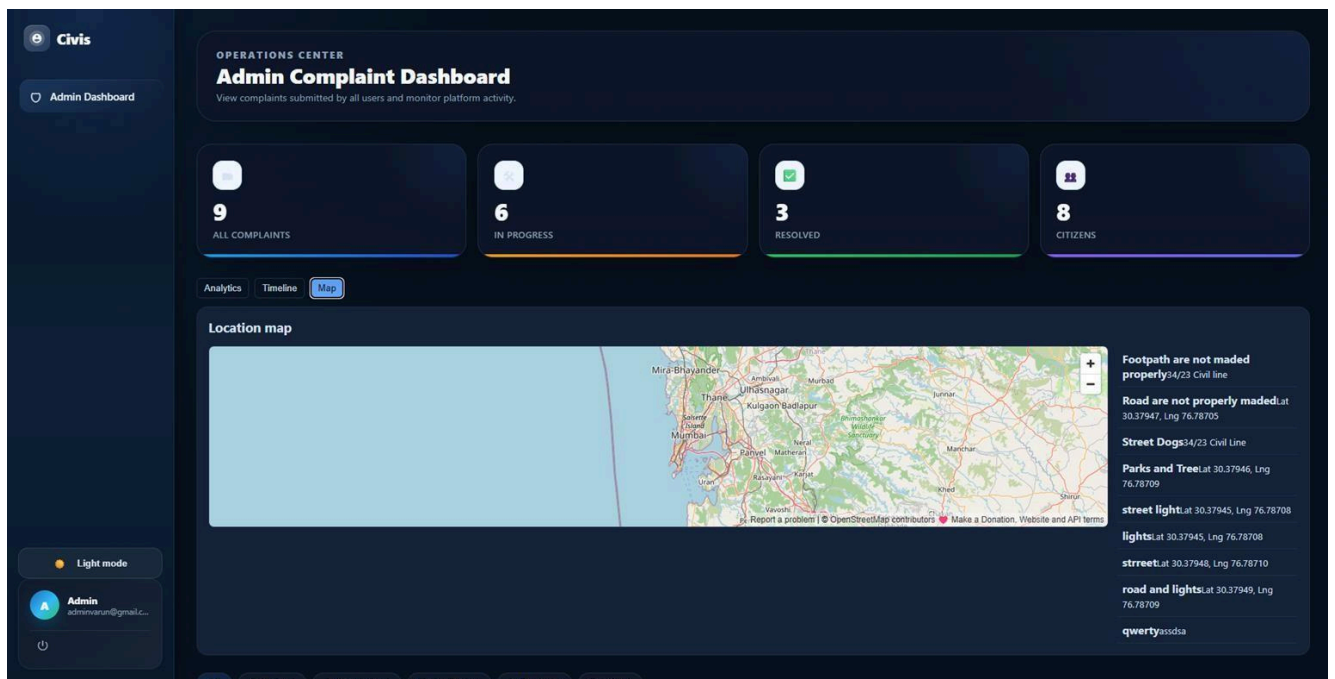
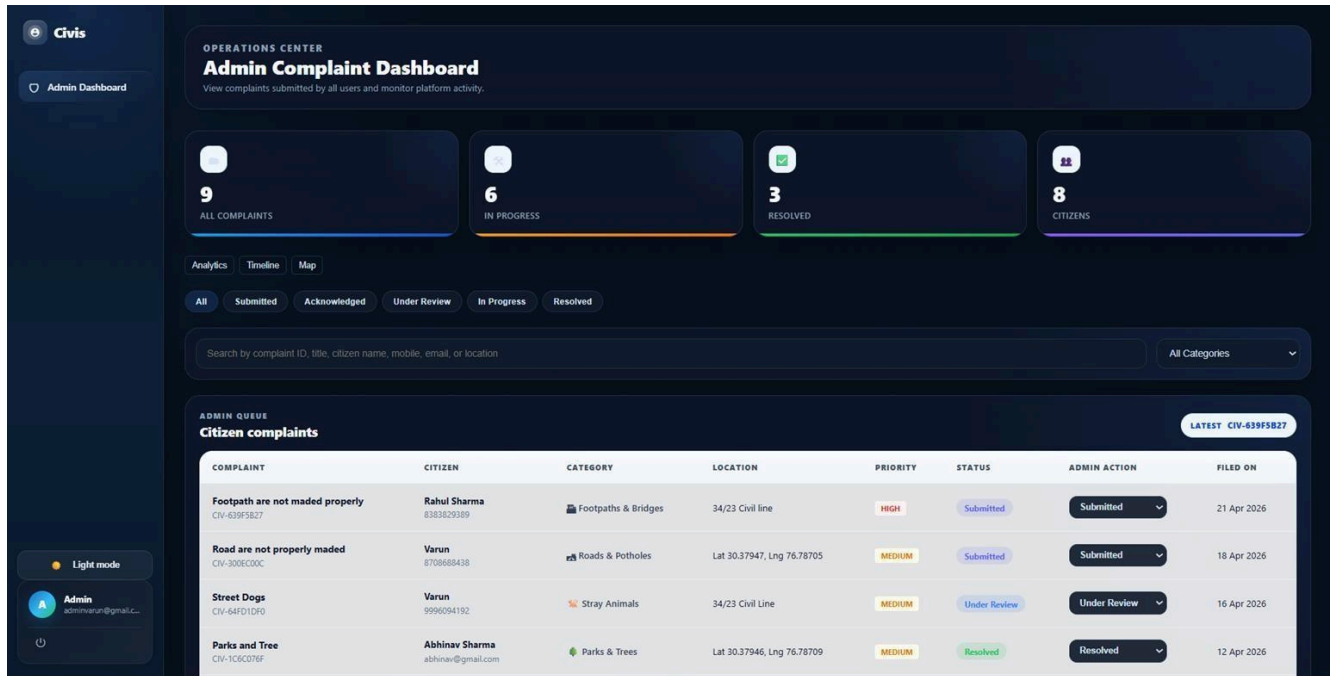




1.6.2 How It Works



1.6.3 Admin Dashboard



1.6.4 File New Complaints

Civis

Category

Details

Location

Review

Where is the issue?

Step 2 of 4

AREA / STREET / WARD *
e.g. Sector 14, MG Road, Ward 7

LANDMARK (OPTIONAL)
e.g. Near HDFC Bank, behind City Mall

Pick the issue on the map
Click anywhere on the map to drop a pin, or use your current location.

Use my location

No map pin selected yet.

Back

Review

Clear complaints with exact location get resolved 2x faster

1.6.5 Review Complaints

Dashboard

CV-4F34E3C

ROADS & POTHOLES

lights

Filed on 9 April 2026 - Lat 30.37943, Long 76.76708

Submitted

HIGH PRIORITY

COMPLAINT STATUS TRACKER

1 Submitted
Current stage

2 Acknowledged

3 Under Review

4 In Progress

5 Resolved

DESCRIPTION

Issue reported: lights. This problem is affecting nearby residents and needs attention from the concerned civic department.

ACTIVITY TIMELINE

Submitted

Your complaint has been received and is under processing.

9 Apr 2026, 12:40 pm

File Another Complaint

Back to Dashboard

1.7 Original Contribution Statement

The main purpose behind this is creating a unified digital platform that causes problems between public and officials regarding civic issues. The system consolidates key functionalities - such as complaint submission, complaint tracking, administrative review and status history maintenance - into a single, well-structured application.

One of the major contributions of this is to provide a clear and visible complaint handling process. By offering citizens the ability to view real time information and trace every stage of their complaint, the system promotes accountability and make sure everything process is saved in record

From the technical perspective, the project shows the real-life example of how it is used for modern civic complaint management solutions using up-to-date online development tools. It shows how an online system can simplify public issues reporting, make people work better together between administrative departments and improve communication flow between people and officials.

Overall, this project advances the idea of work better and responds faster to civic service management by delivering a reliable, structured and user - focused system for complaint registration, monitoring and resolution.